

Derek Corniello

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EDUCATION

University of Cincinnati, Cincinnati, Ohio

Expected Graduation: May 2027

Bachelor of Science, Computer Science

GPA: 3.9

Master of Engineering, Software Engineering

GPA: 4.0

EXPERIENCE

LinkedIn | Software Engineer Intern | Mountain View, CA

May 2026 - August 2026

- Led development of workflow translation engine that converted business-authored definitions into executable workflows
- Streamlined authoring with a declarative YAML layer, reducing end-user effort by 50% via the translation engine
- Modernized 5 AI platform's microservice integrations by upgrading to gRPC with both streaming and unary capabilities
- Established platform standards and workflows by authoring canonical technical and architectural development standards

Fifth Third Bank | Software Engineer Intern | Cincinnati, OH

January 2026 - May 2026

- Contributed to the modernization of an IVR system, migrating from a telecommunications platform to AWS infrastructure
- Developed 20+ backend REST API microservice components to support over 100,000 daily callers and customer data flow
- Collaborated on changes across 5 different domains, debugging 30+ full-stack services supporting system modernization
- Participated in code reviews and cross-team collaboration, improving code quality and stability in a high-traffic system

Siemens Digital Industries Software | Software Engineering Intern | Milford, OH

January 2024 - December 2025

- Collaborated with 3 cross-functional teams to design, develop, and test over 10 innovative software solutions
- Built REST APIs with sub-millisecond responses, operating on 100,000+ database entries efficiently on dozens of endpoints
- Implemented AI features using RAG, embeddings, graphs and ETL pipelines, enhancing LLM efficiency and accuracy
- Administered over 200 pages of documentation of projects, facilitating seamless knowledge transfer and improving SDLC

PROJECTS

Mux Programming Language | mux-lang.dev | Rust, LLVM

- A strong, statically-typed, compiled programming language designed to help programmers write simple, clean, and easy code
- Built core compiler pipeline including lexing, parsing, semantic analysis, and intermediate representation generation
- Created a runtime and standard library foundation to support core language functionality and developer usability

hunch | [GitHub](https://github.com) | Go, zsh, SQLite

- Designed a CLI predictor that learns workflow patterns from command history using a Markov-style transition graph
- Implemented a two-phase command normalization pipeline to generalize raw shell commands into reusable templates
- Architected a daemon and client IPC system over Unix domain sockets, enabling cross-platform sub-millisecond latency

SKILLS

Languages: Python, Go, Rust, Java, TypeScript, C++, SQL

Development Tools: git, neovim, AWS, docker, Linux, Unity

LEADERSHIP AND CERTIFICATIONS

Amazon Web Services (AWS) Student Builder Center Leader

August 2026 - Present

Association for Computing Machinery (ACM) - Executive Officer

August 2026 - Present

Best Computer Science Undergraduate Student

April 2026

Amazon Web Services (AWS) Certified Cloud Practitioner

February 2026

Institute of Electrical and Electronics Engineers: Eta Kappa Nu (IEEE-HKN) Honors - Executive Officer

October 2025 - May 2026